Linear Wave Resistance of an Image-Inspired Displacement Yacht

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The wave resistance of displacement sailing yachts changes rapidly when the Froude number brings the bow- and stern-generated waves into and out of phase. This paper assesses whether a transparent linear potential-flow method can resolve this variation for a broad yacht form at early-design cost. The test case is a generic, fixed-attitude, 10 m hull reconstructed from an undimensioned lines-plan image, with a beam-to-length ratio of 0.28 and a draft-to-length ratio of 0.06. A three-dimensional Havelock source distribution is integrated through a Kochin far-field formulation over Froude numbers from 0.15 to 0.45. The wave-resistance coefficient increases from 0.00195 at the lowest speed to a maximum of 0.03859 at a Froude number of 0.425, which corresponds to 7.61 kN for the selected scale. The maximum difference between the fine and medium hull grids is 1.71% for Froude numbers greater than or equal to 0.20, while the far-field quadrature tail remains below 5.10×10^{-4}. An independent thin-Wigley benchmark differs from Michell's analytic result by at most 5.61%. Far-field maps reveal the associated transition from short, densely spaced wave crests to longer, more transverse-dominated patterns. The method therefore resolves useful speed-dependent trends, but its broad-hull predictions require validation before quantitative design use.

Key words: sailing yacht, ship waves, wave resistance, potential flow, Kochin function, hull-form screening

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Nomenclature

a	complex Kochin amplitude
C_W	wave-resistance coefficient
Fn	Froude number
g	gravitational acceleration
I	Kochin resistance integral
L	reference waterline length
n_x	streamwise component of the hull-panel normal
R_W	wave resistance
S	wetted surface area
t	transverse spectral coordinate
u_∞	magnitude of the free-stream velocity
x, y, z	streamwise, transverse and vertical coordinates
λ	spectral parameter, $\lambda \equiv \sqrt{1+t^2}$
ρ	water density
σ	source strength
ζ	free-surface elevation or wave-pattern field

1. Introduction

Wave making is a principal component of the calm-water resistance of a displacement sailing yacht at moderate and high Froude numbers. It also varies non-monotonically with speed because waves generated by different parts of the hull interfere. Michell's thin-ship theory provides the classical linear description of this process (Michell 1898; Tuck 1989). Havelock subsequently expressed wave resistance in terms of distributions of singularities beneath the free surface (Havelock 1932). These theories remain attractive in early-stage design because a complete speed sweep can be obtained without the cost of a viscous free-surface calculation.

The thin-ship assumption is restrictive for sailing yachts, whose immersed forms may combine a relatively large beam-to-length ratio, rounded sections and pronounced longitudinal rocker. Rankine-source panel methods retain three-dimensional geometry and have therefore become a standard route from classical theory to practical ship-wave computation (Dawson 1977). Discretisation can nevertheless distort the propagation of steady ship waves, and resistance predictions require explicit verification rather than reliance on a converged linear solve alone (Nakos & Sclavounos 1990). Three-dimensional wake spectra have been shown to provide a direct connection between the wave field and the integrated resistance (Nakos & Sclavounos 1994).

The present paper addresses the following question: can a compact, transparent linear source method resolve the speed-dependent resistance and wake topology of a broad displacement-yacht test case with quantified numerical sensitivity? To answer it, a generic hull reconstructed from the supplied lines-plan image is examined over $0.15 \leqslant Fn \leqslant 0.45$. The geometry, formulation and verification procedure are described in §2. The resistance curve, numerical diagnostics, far-field spectra and wave patterns are presented in §3. The implications and limitations are discussed in §4, followed by the conclusions in §5.

2. Methodology

The numerical test is designed to isolate wave making. The yacht advances at constant speed in calm, inviscid, incompressible water of infinite depth. Its attitude is fixed; sinkage and trim are zero. Appendages and the above-water geometry are omitted.

Dimensional variables are denoted by a hat in this section. All reported coordinates are nondimensionalised by the waterline length $\hat{L}$, velocities by the free-stream speed $\hat{u}_\infty$ and areas by $\hat{L}^2$.

2.1. *Hull geometry and test matrix*

The body-fixed frame $\mathrm{O}(x, y, z)$ has x increasing from bow to stern, y positive to starboard and z positive upwards. The undisturbed free surface is $z = 0$. The hull is a parametric reconstruction of the principal immersed forms visible in the supplied raster lines plan. The breadth and draft vary smoothly between pointed ends, while superelliptic sections reproduce rounded U-shaped midship sections and increasingly V-shaped forward sections (figure 1). Because the source image has no common dimensional scale, the ratios $B/L = 0.28$ and $T/L = 0.06$ are imposed independently. The geometry is therefore a generic displacement yacht rather than a reproduction of a Dufour 39.

The dimensional reference length is $\hat{L} = 10\,\mathrm{m}$. The maximum beam and canoe-body draft are 2.799 m and 0.600 m, respectively. Numerical integration of the offsets gives a displaced volume of $7.111\,\mathrm{m}^3$, a seawater displacement of 7289 kg, a wetted area of $21.760\,\mathrm{m}^2$ and a waterplane area of $17.948\,\mathrm{m}^2$. The longitudinal centre of buoyancy is 0.395 m aft of midships. Seawater density is $\hat{\rho} = 1025\,\mathrm{kg\,m^{-3}}$ and gravitational acceleration is $\hat{g} = 9.806\,65\,\mathrm{m\,s^{-2}}$.

The Froude number is

$$Fn \equiv \frac{\hat{u}_\infty}{\sqrt{\hat{g}\hat{L}}}. \tag{2.1}$$

Thirteen operating points are evaluated at increments of 0.025 over $0.15 \leqslant Fn \leqslant 0.45$, corresponding to speeds from $1.486\,\mathrm{m\,s^{-1}}$ to $4.456\,\mathrm{m\,s^{-1}}$. Six operating points, $Fn = 0.20$, 0.25, 0.30, 0.35, 0.40 and 0.45, are selected for wave-pattern analysis.

2.2. *Linear potential-flow formulation*

The disturbance potential ϕ satisfies Laplace's equation. On the mean hull surface and mean free surface, respectively, the linear boundary conditions are

$$\boldsymbol{\nabla}\phi \boldsymbol{\cdot} \boldsymbol{n} = -n_x, \qquad Fn^2 \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial \phi}{\partial z} = 0 \quad (z = 0), \tag{2.2}$$

with an upstream radiation condition. The robust resistance path uses the linearised Havelock source distribution $\sigma_j = -n_{x,j}$ on each actual three-dimensional hull panel. Hence, finite breadth, depth, area and transverse phase are retained, although the source strength is linearised.

For the transverse spectral coordinate t, the wavenumber components are

$$\lambda \equiv \sqrt{1 + t^2}, \qquad k_x \equiv \frac{\lambda}{Fn^2}, \qquad k_y \equiv \frac{\lambda t}{Fn^2}, \qquad k \equiv \frac{\lambda^2}{Fn^2}. \tag{2.3}$$

Analytical reflection of the starboard half hull gives the Kochin amplitude

$$a(t) \equiv -\sum_j \sigma_j \mathit{\Delta} S_j \exp\left(k z_j - \mathrm{i} k_x x_j\right) \cos(k_y y_j), \tag{2.4}$$

where $\mathrm{i} \equiv \sqrt{-1}$. The positive-definite resistance integral and the wetted-area-normalised wave-resistance coefficient are

$$I \equiv \int_0^\infty \lambda |a(t)|^2 \,\mathrm{d}t, \qquad C_\mathrm{W} \equiv \frac{8I}{\pi Fn^4 (S/L^2)}. \tag{2.5}$$

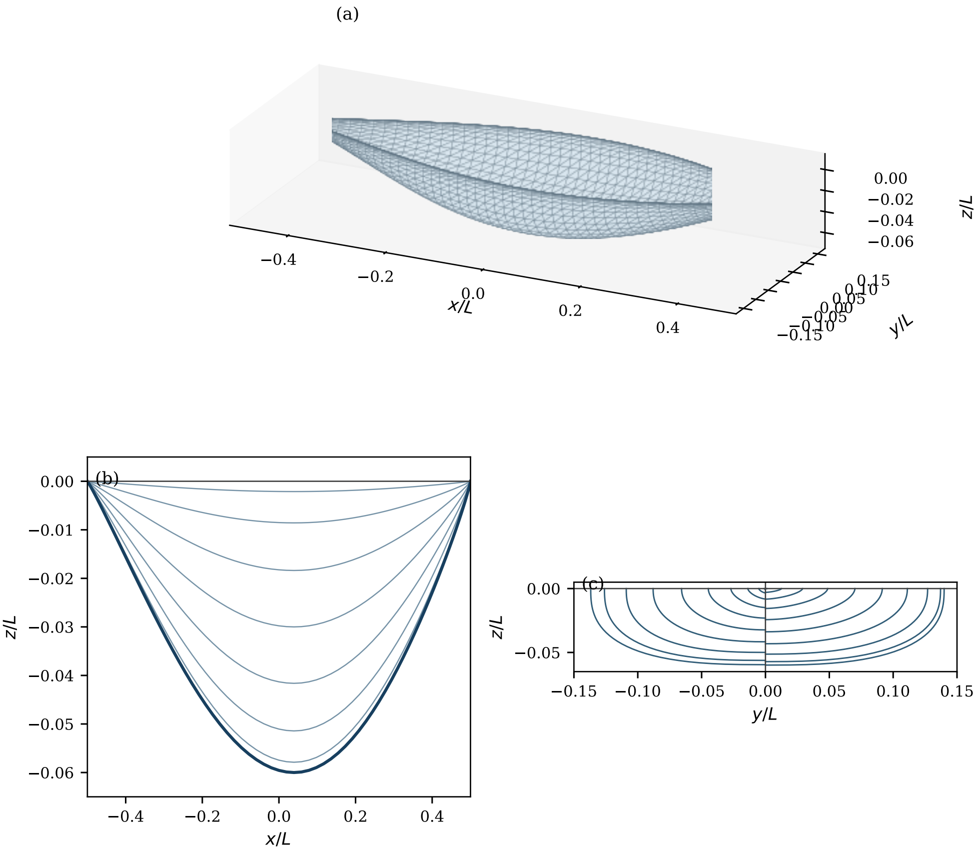

FIGURE 1. Image-inspired displacement-yacht geometry: (a) fine triangular surface mesh, (b) profile and longitudinal section lines, and (c) body plan with aft sections on the left and forward sections on the right. The waterline is $z/L = 0$.

This normalisation is equivalent to

$$\hat{R}_{\mathrm{W}} \equiv \frac{1}{2}\hat{\rho}\hat{u}_{\infty}^{2}\hat{S}C_{\mathrm{W}}. \tag{2.6}$$

The integral is evaluated on 6001 quadrature points after the transformation $t = q^2$, with $0 \leqslant t \leqslant 60$. The contribution from the final 20% of the interval divided by I is reported as the quadrature-tail indicator. The full implementation and machine-readable outputs are distributed with the article (Viola 2026).

2.3. *Far-field wave-pattern reconstruction*

The Kochin spectrum is also used to visualise the radiating far field. This avoids presenting the compact free-surface-panel solution as a converged local elevation. The plotted field is

$$\zeta_{\mathrm{N}}(x, y) \equiv \frac{\mathrm{Re}\left\{\int_0^{12} w(t)\lambda a(t)\exp(\mathrm{i}k_x x)\cos(k_y y)\,\mathrm{d}t\right\}}{\max\limits_{x,y}\left|\mathrm{Re}\left\{\displaystyle\int_0^{12} w(t)\lambda a(t)\exp(\mathrm{i}k_x x)\cos(k_y y)\,\mathrm{d}t\right\}\right|}, \tag{2.7}$$

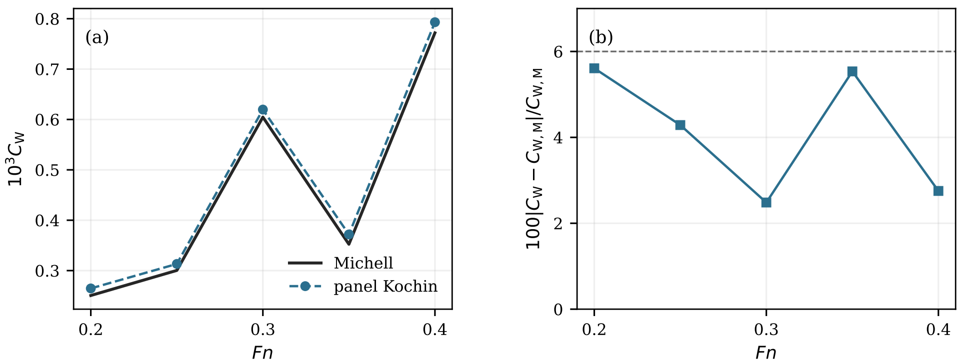

FIGURE 2. Wigley thin-limit verification: (a) wave-resistance coefficient from the panel Kochin integral and independent Michell evaluation; (b) absolute relative difference, with the dashed line marking 6%. Here $B/L = 0.05$ and $T/L = 0.0625$.

where $w = 1$ for $t \leqslant 9.6$ and decreases to zero through a raised-cosine taper at $t = 12$. Each operating point is normalised independently. Consequently, equation (2.7) shows phase, wavelength and interference topology, but not the variation of dimensional wave amplitude between speeds. Wave-pattern resistance inferred from measured longitudinal cuts is a useful independent validation quantity (International Towing Tank Conference 2021); no such measurements are available for the present synthetic hull.

2.4. *Solution verification and numerical sensitivity*

Implementation verification uses a Wigley hull with $B/L = 0.05$ and $T/L = 0.0625$. Its slenderness permits an independent analytic evaluation of Michell's integral. The 41 by 15 point panel solution differs from the analytic coefficient by 2.50–5.61% over $0.20 \leqslant Fn \leqslant 0.40$ (figure 2). This comparison verifies the normalisation, surface integration and spectral quadrature in the limit for which the reference theory applies; it does not validate the broad yacht case.

The yacht is discretised with coarse, medium and fine half-hull grids of 31 by 13, 41 by 15 and 51 by 19 points. Degenerate triangles at the stem, stern and centreline are removed, leaving 696, 1092 and 1764 active panels. The absolute fine-to-medium difference divided by the fine result is used as a discretisation-sensitivity estimate. A formal grid convergence index is not reported because the two surface directions do not have one common refinement factor and the lowest-speed response is not monotonically converged. This choice separates a directly observed sensitivity from a statistical or asymptotic uncertainty estimate (Eça & Hoekstra 2014).

3. Results

3.1. *Wave resistance*

The three grids give visually coincident resistance curves over most of the speed range (figure 3a). The coefficient increases from 0.001951 at $Fn = 0.15$ to 0.007616 at $Fn = 0.275$, decreases slightly to 0.007586 at $Fn = 0.30$, and then rises steeply. It reaches a maximum of 0.038591 at $Fn = 0.425$ before decreasing to 0.037686 at $Fn = 0.45$. The hollow near $Fn = 0.30$ and the high-speed crest are consistent with destructive and constructive interference between longitudinally separated source regions.

At the selected scale, the maximum coefficient occurs at $\hat{u}_\infty = 4.209\,\mathrm{m\,s^{-1}}$ and gives $\hat{R}_W = 7.613\,\mathrm{kN}$ (figure 3b). The dimensional resistance continues to increase at

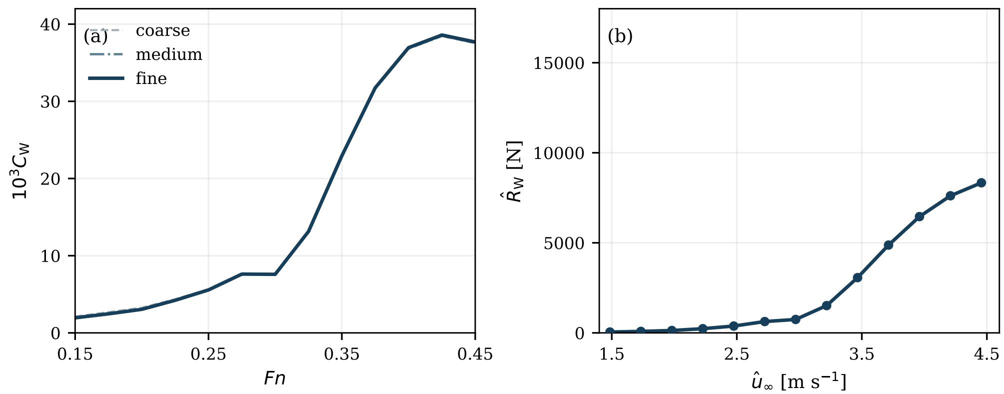

FIGURE 3. Wave resistance versus speed: (a) area-normalised coefficient for the three hull grids; (b) dimensional resistance for the fine grid and the 10 m test case in seawater.

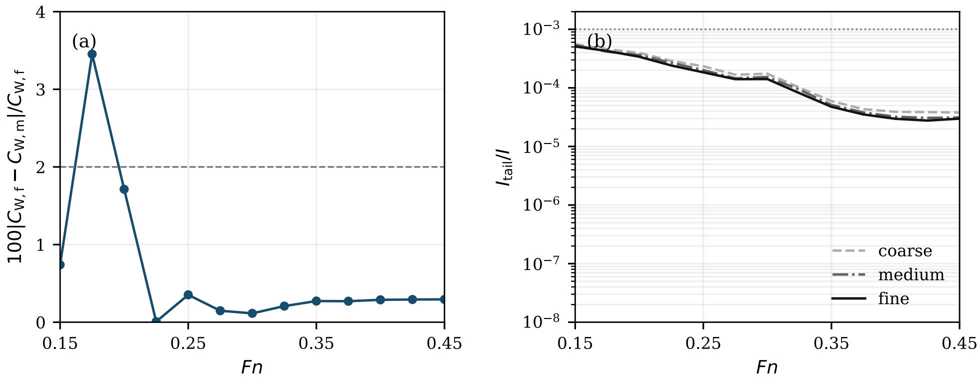

FIGURE 4. Numerical diagnostics versus Froude number: (a) absolute fine-to-medium relative difference in wave-resistance coefficient; (b) final-20% Kochin quadrature contribution for each grid, with the dotted line marking $I_{\mathrm{tail}}/I = 10^{-3}$.

$Fn = 0.45$, despite the slight decrease in coefficient, because the dynamic pressure is proportional to speed squared. This result is wave resistance alone and must not be interpreted as the total calm-water resistance.

3.2. *Numerical diagnostics*

The fine-to-medium difference is below 1.71% for every case with $Fn \geqslant 0.20$, and its mean is 0.36% over this subset (figure 4a). The difference rises to 3.45% at $Fn = 0.15$, where the shorter waves demand greater longitudinal resolution. The reported curve should therefore be considered less resolved at the lower boundary of the declared envelope.

The quadrature tail remains below 5.10×10^{-4} for the fine grid and below 5.55×10^{-4} across all three grids (figure 4b). The largest values occur at the two lowest Froude numbers. These indicators show that the finite Kochin interval contains the computed spectrum to the prescribed tolerance, independently of the hull grid comparison.

3.3. *Kochin spectra and wave patterns*

The normalised integrand $\lambda|a|^2$ changes systematically with speed (figure 5). At the lower Froude numbers, several comparable peaks extend over the resolved transverse

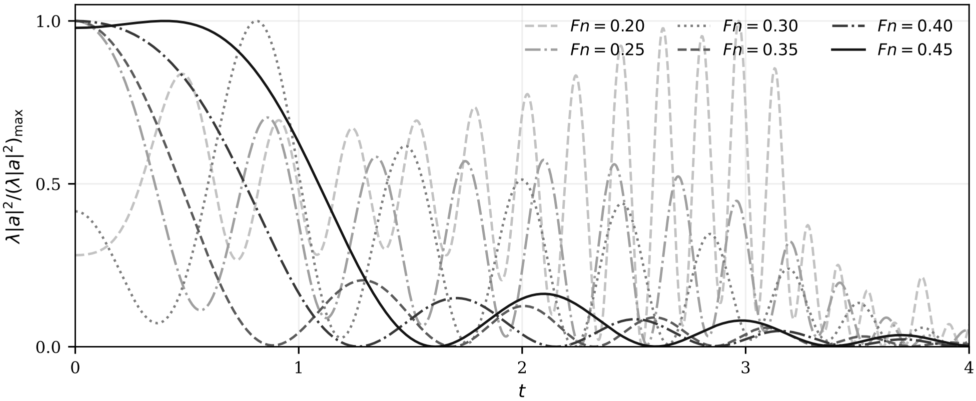

FIGURE 5. Normalised Kochin resistance-integrand density versus transverse spectral coordinate for six Froude numbers. Each curve is divided by its own maximum to expose changes in spectral distribution.

spectrum. Increasing the Froude number concentrates the dominant contribution towards smaller t, while secondary lobes retain the signature of bow–stern interference. The concentration is particularly strong at $Fn = 0.40$ and 0.45, where the dominant low-t content coincides with the crest of the resistance curve.

The far-field maps translate these spectra into physical-space topology (figure 6). At $Fn = 0.20$, the wake contains closely spaced, high-angle divergent crests. The characteristic spacing increases from $Fn = 0.25$ to 0.30, while alternating bands reveal interference between the forward and aft source regions. At $Fn = 0.35$, a stronger long-wave component appears immediately downstream and extends into coherent oblique arms. The $Fn = 0.40$ and 0.45 patterns are dominated by longer waves and broader transverse bands, consistent with their low-t spectra.

The maps are symmetric about the centreplane because only upright, zero-leeway motion is considered. They remain confined to the classical radiating wake and show no breaking, viscous attenuation or near-field crest steepening. The visual progression nevertheless explains why a scalar resistance curve alone is insufficient: similar coefficients can result from materially different distributions of wave direction and phase.

4. Discussion

The test supports the thesis that a low-order three-dimensional source method can resolve useful speed-dependent trends for a displacement-yacht form. The positive Kochin integral prevents a non-physical negative resistance, the three-grid comparison bounds the observed discretisation sensitivity, and the far-field reconstruction provides a direct qualitative explanation of the humps and hollow. A complete thirteen-point fine-grid sweep, including figures and machine-readable output, is reproducible with one Python script.

The numerical evidence has clear boundaries. First, the imposed source strength is a linear Neumann–Kelvin approximation. The test hull has $B/L = 0.28$ and is therefore substantially broader than the Wigley verification body. The 5.61% benchmark error must not be transferred as an uncertainty for the yacht. Secondly, the fixed-attitude model omits dynamic sinkage and trim, which can shift interference features. Thirdly, viscosity, boundary-layer growth, separation, appendage drag, wave breaking

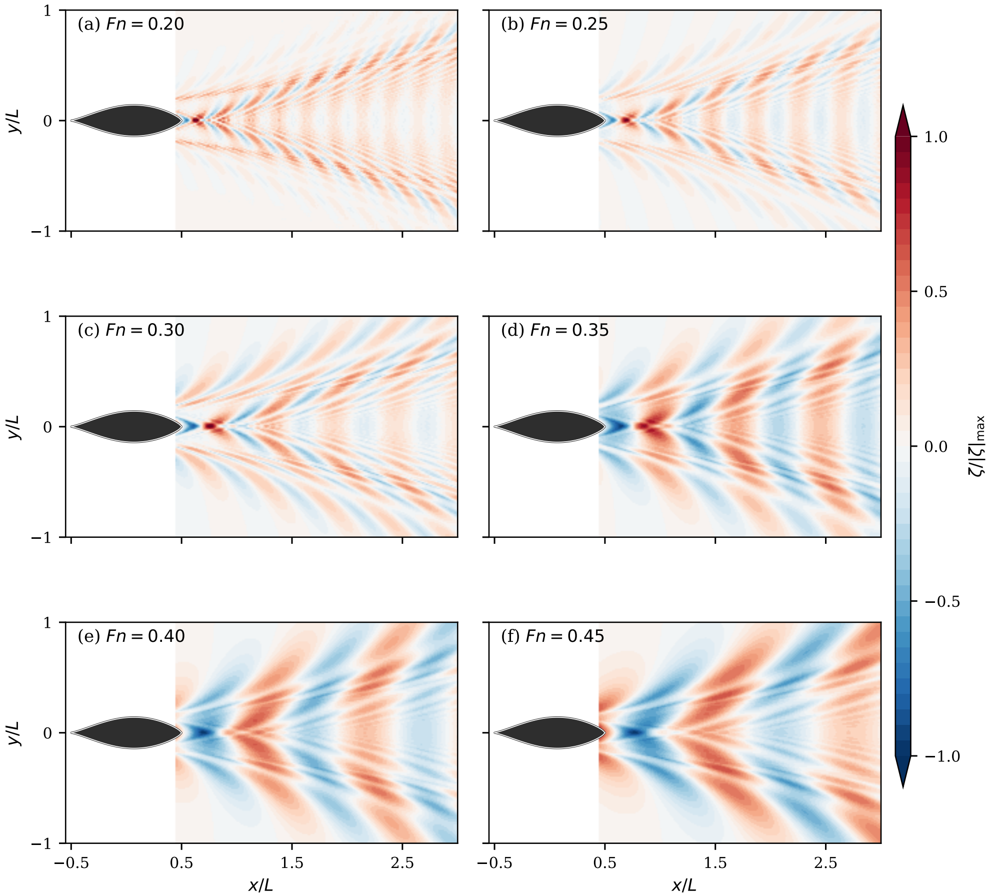

FIGURE 6. Normalised far-field wave-pattern phase for (a) $Fn = 0.20$, (b) 0.25, (c) 0.30, (d) 0.35, (e) 0.40 and (f) 0.45. Red and blue denote opposite phases; each field is independently normalised and therefore does not compare wave amplitudes between Froude numbers. Flow and vessel motion are towards positive x.

and transom ventilation are absent. Finally, the plotted wave fields are independently normalised far-field phase reconstructions rather than dimensional near-field elevations.

The resistance level, especially the high-speed coefficient, should therefore be interpreted as a screening prediction. Quantitative design use requires a like-for-like towing-tank or validated viscous free-surface computation. The most informative experiment would measure both resistance and longitudinal wave cuts at the same fixed attitude. That combination permits direct comparison of the scalar force, the wave spectrum and the spatial pattern in accordance with the International Towing Tank Conference procedure (International Towing Tank Conference 2021).

5. Conclusions

A linear potential-flow test has been performed for a broad, image-inspired displacement yacht over Froude numbers from 0.15 to 0.45. The method represents the three-dimensional immersed surface with sources and obtains wave resistance from a positive far-field energy integral. A ten-metre hull in seawater provides the dimensional reference.

The calculation resolves a shallow resistance hollow near a Froude number of 0.30

followed by a pronounced high-speed rise. The wave-resistance coefficient reaches 0.03859 at a Froude number of 0.425, corresponding to 7.61 kilonewtons at the selected scale. Medium-to-fine grid differences remain below 1.71 per cent from a Froude number of 0.20 upwards, and the unresolved tail of the far-field integral remains below 0.051 per cent. The associated wave patterns change from densely spaced divergent crests to longer patterns dominated by broad transverse bands.

These results demonstrate that the method is suitable for rapid, internally consistent comparisons of displacement-yacht hull forms and operating speeds. They do not establish predictive accuracy for a real yacht. The principal limitations are the linearised source strength, fixed attitude, inviscid flow and absence of experimental validation. The next step is a controlled comparison with resistance and wave-cut measurements for a fully specified hull geometry.

Acknowledgements

The computational test uses the hull-form image supplied for this project. No external experimental data were used.

Declaration of interests

The Author reports no conflict of interest.

Data availability statement

The source code, exact test-case script, tabulated resistance curve, wave fields and all figure sources are available in the project repository. The calculation is reproduced from the repository root with `python examples/journal_test_case.py`; the manuscript is compiled with `latexmk -pdf article/main.tex`.

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